Manual Calculation

Exp - 1

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① program A: 8-bit Addition & subtraction

given A = 5AH, B = 27H

Addition:

5AH + 27H = 81H

Sum = 81H, CY = 0

AC = 1, P = 1, Z = 0, S = 1

Flag byte = 94H

signed overflow = yes

Subtraction:

5AH - 27H = 33H

Difference = 33H, Borrow = 0

AC = 0, P = 1, Z = 0, S = 0

Flag byte = 06H

signed overflow = NO

② program B: 16-bit Addition and subtraction

given X = 1234H, Y = 4321H

Addition:

1234H + 4321H = 5555H

Carry = 0

subtraction:

1234H - 4321H = CF13H

Borrow = 1

③ program C: 8-bit Multiplication and division

given Multiplication:

14H × 06H = 0078H

Product = 0078H

Division:

F2H ÷ 0AH

Quotient = 18H

Remainder = 02H

Verification:

F2H = (18H × 0AH) + 02H

Program D: 16-bit Multiplication and division

given

Multiplication:

1357H × 0025H = 0002CA13H

Upper word = 0002H

Lower word = CA13H

Division:

9A23H ÷ 0011H

Quotient = 0911H

Remainder = 0002H

Verification:

9A23H = (0911H × 0011H) + 0002H